Lecture 7

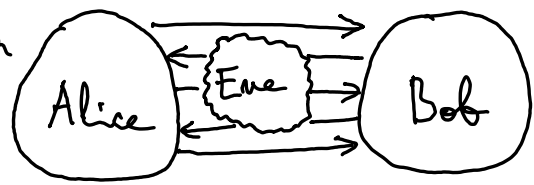
This lecture is on applications of quantum information to communication and cryptography. Exchanging information encoded in quantum states rather than bits can have surprising advantages. Also, building two small quantum computers that can communicate and perform simple operations on one or two qubits can be easier than making one large quantum computer that can store many qubits and run complicated algorithms.

We call algorithms that involve several parties who exchange information, protocols. The three most important protocols in quantum computing are:

- quantum key distribution (QKD) ← We will not cover this.
- superdense coding
- quantum teleportation

They are not about achieving computational speed-up but instead about exploiting the unusual properties of quantum information, such as superposition and entanglement.

A typical scenario involves two friends, Alice and Bob, who want to jointly perform some task, such as exchange information, but they want the information to remain private from eavesdropper Eve who tries to intercept their messages:



One-time pad

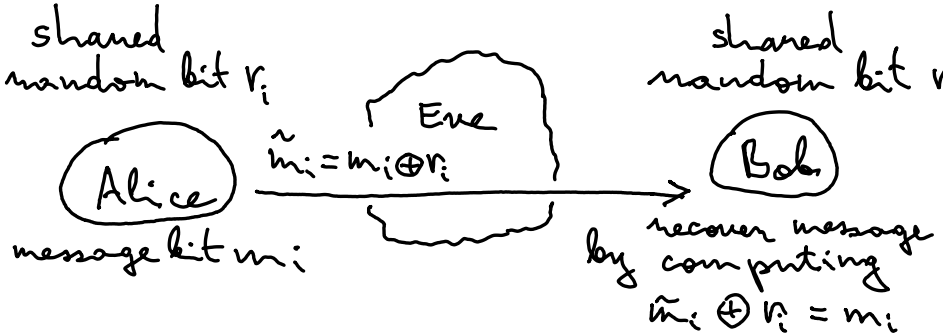
Is there in principle a perfectly secure way for Alice and Bob to communicate? yes!

Goal: Send a private message from Alice to Bob using public communication.

Protocol:

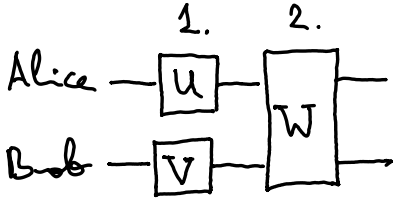
1. Preparation: Alice and Bob meet upfront to generate n shared uniformly random bits $r_1, r_2, \ldots, r_n$ where n is the length of the message they want to exchange. Then they part.
2. Encoding: If the i-th message bit Alice wants to send is m_i, she sends $m_i \oplus r_i$.
3. Decoding: If Bob receives $\tilde{m}_i$, he can find the actual message bit by computing

$$\tilde{m}_i \oplus r_i = (m_i \oplus r_i) \oplus r_i = m_i$$



Security: If Eve intercepts $\tilde{m}_i$, she gains no information about the actual message m_i since $\tilde{m}_i$ appears uniformly random to her. This is called information-theoretic security. (No amount of computational power can help Eve to recover m_i.)

Local vs global operations



1. Local operations: Alice applies U and Bob applies V on their respective qubits. The overall operation is $U \otimes V$.

2. Global operation: Alice and Bob put their qubits together and jointly apply W.

Bell states

The following states form an orthonormal basis for two qubits:

The EPR (Einstein-Podolsky-Rosen) state

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \qquad |\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

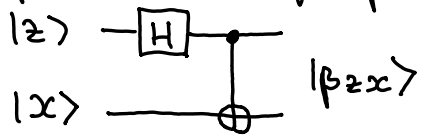
$$|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \qquad |\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

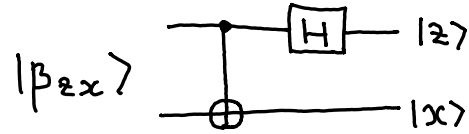
More concisely, for $z, x \in \{0,1\}$,

$$|\beta_{zx}\rangle = \frac{1}{\sqrt{2}}\left(|0, x\rangle + (-1)^z |1, x \oplus 1\rangle\right)$$

Recall that $H|z\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^z|1\rangle)$, for all $z \in \{0,1\}$, and $\text{CNOT}\,|a, x\rangle = |a, x \oplus a\rangle$, for all $a, x \in \{0,1\}$.

Preparation / unpreparation of $|\beta_{zx}\rangle$:





$$|\beta_{zx}\rangle = \text{CNOT} \cdot (H \otimes I) \cdot |z, x\rangle \qquad |z, x\rangle = (H \otimes I) \cdot \text{CNOT} \cdot |\beta_{zx}\rangle$$

Note that CNOT is a global operation on both qubits.

Local conversion: surprisingly, one can convert any Bell state into any other by acting only on one of the qubits, i.e., by a local operation:

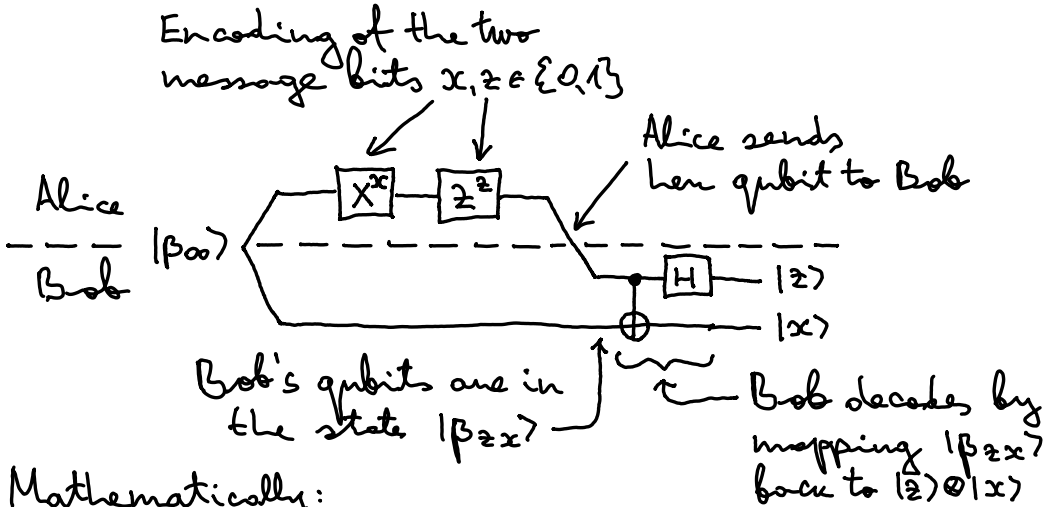
$$|\beta_{zx}\rangle = (Z^z X^x \otimes I)\cdot|\beta_{00}\rangle = (I \otimes X^x Z^z)\cdot|\beta_{00}\rangle$$

These local operations are just a combination of Pauli X and Z matrices.

Superdense coding

Goal: send 2 classical bits by transmitting 1 qubit.

This can be achieved by using the local conversion and orthogonality of Bell states:



Mathematically:

$$\boxed{(H\otimes I)\cdot CNOT\cdot(Z^z X^x\otimes I)\cdot|\beta_{00}\rangle = |z,x\rangle}$$

Resource trade-off:

1 shared EPR state + 1 qubit of quantum communication = 2 bits of classical communication

One-time pad with probabilistic bits

With one-time pad, Alice can in fact send even a probabilistic bit $\binom{p_0}{p_1}$ by transmitting what appears to Eve as a uniformly random bit.
Let us write $[0]$ and $[00]$ as the probabilistic versions of the computational basis states $|0\rangle$ and $|00\rangle$.
Let

$$m = p_0[0] + p_1[1] = \begin{pmatrix} p_0 \\ p_1 \end{pmatrix}.$$

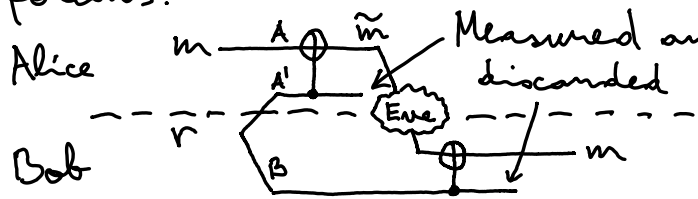
be the probabilistic bit Alice wants to send to Bob. The shared uniformly random bit is then

$$r = \tfrac{1}{2}[00] + \tfrac{1}{2}[11] = \tfrac{1}{2}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

All three bits together look as follows:

$$m \otimes r = \left(p_0[0] + p_1[1]\right)_A \otimes \left(\tfrac{1}{2}[00] + \tfrac{1}{2}[11]\right)_{A'B}$$

where Alice has the first two bits A and A' and Bob has the third bit B. The one-time pad protocol looks as follows:



After the first CNOT: $\frac{1}{2}\binom{p_0}{p_1}_A \otimes [00]_{A'B} + \frac{1}{2}\binom{p_1}{p_0} \otimes [11]$.

Discarding A': $\frac{1}{2}\binom{p_0}{p_1}_A \otimes [0]_B + \frac{1}{2}\binom{p_1}{p_0} \otimes [1]$. Eve sees only system A which to her looks like $\frac{1}{2}\binom{p_0}{p_1} + \frac{1}{2}\binom{p_1}{p_0} = \frac{1}{2}\binom{p_0+p_1}{p_1+p_0} = \frac{1}{2}\binom{1}{1}$, i.e., uniformly random. After his CNOT, Bob gets:

$$\frac{1}{2}\binom{p_0}{p_1} \otimes [0] + \frac{1}{2}\binom{p_0}{p_1} \otimes [1] = m \otimes \frac{1}{2}\binom{1}{1}.$$

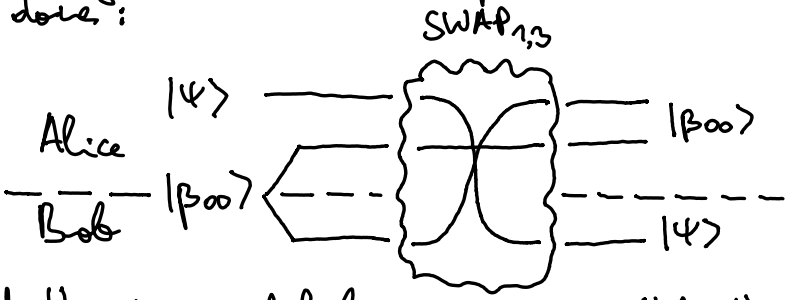
After discarding the second bit he recovers the original message m.

Quantum teleportation

This is the quantum version of the one-time pad - instead of transmitting a probabilistic bit, Alice transmits a quantum bit!

One-time pad	Quantum teleportation
1 shared random bit	1 shared EPR state
+	+
1 bit of public communication	2 bits of public communication
=	=
1 probabilistic (private) bit of communication	1 qubit of quantum communication

This is how you could try to invent the protocol... If only we could SWAP qubits 1 and 3, we would be done:



But this is a global operation. Note that Pauli matrices form a basis for all 2×2 complex matrices and, similarly, tensor products of pairs of Pauli matrices forms a basis for all 4×4 matrices, so we can expand SWAP as a linear combination of local (Pauli) operations.

Recall that

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

You can check that

$$\text{SWAP} = \frac{1}{2}\left(I \otimes I + X \otimes X + Y \otimes Y + Z \otimes Z\right)$$

$$= \frac{1}{2} \sum_{z,x \in \{0,1\}} Z^z X^x \otimes X^x Z^z.$$

Let's apply this to the 1st and 3rd qubit:

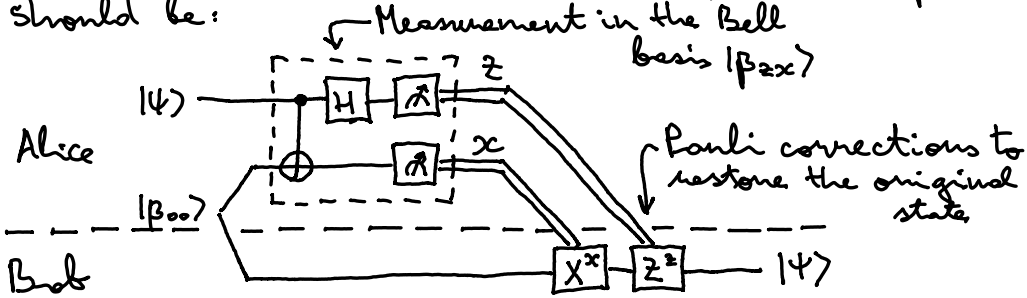
$$|\psi\rangle \otimes |\beta_{00}\rangle = \text{SWAP}_{1,3} \cdot |\beta_{00}\rangle \otimes |\psi\rangle$$

$$= \frac{1}{2} \sum_{z,x \in \{0,1\}} \underbrace{(Z^z X^x \otimes I)|\beta_{00}\rangle}_{|\beta_{zx}\rangle} \otimes X^x Z^z |\psi\rangle$$

We get the teleportation equation:

$$\boxed{|\psi\rangle \otimes |\beta_{00}\rangle = \frac{1}{2} \sum_{z,x \in \{0,1\}} |\beta_{zx}\rangle \otimes X^x Z^z |\psi\rangle}$$

From this we can see what the teleportation protocol should be:



Resource trade-off: 1 shared EPR state
+ 2 bits of classical communication
= 1 qubit of quantum communication